Deepak Mallampati

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Kent, OH - US-based, F-1 OPT, US-Remote ready

PROFESSIONAL SUMMARY

May 2025 Master's grad (Kent State, Computer Science) shipping production healthcare AI at Progress Solutions in HIPAA-conscious environments - agentic LLM workflows (~25% stability), RAG (~35% retrieval precision), predictive ML (15-20% recall on high-risk patient categories), and FastAPI/Docker services with structured logging (~30% post-deploy defect reduction). Eight shipped/ongoing AI projects across Python, TypeScript, React, and SQL. Direct PBM-adjacent healthcare domain context: Patient Records app (Bottle + SQLite RDBMS with normalized patients/diagnoses schema), Agentic Healthcare Claims (5-stage agent pipeline with schema-validated JSON contracts and RAG-grounded CPT/ICD validation against vector-indexed policy documents), and Progress care-workflow + eligibility-check automation. Strong typed-language fundamentals (C++/Arduino + TypeScript) ready to ramp on Golang. CI/CD discipline from Jenkins-based SQL release pipelines (~30% deployment error reduction at Energy Solutions). GenAI tools active in dev workflow (Claude / GPT / Ollama).

CORE COMPETENCIES

Healthcare / PBM Domain (HIPAA-conscious) • RDBMS Query Tuning (T-SQL / PostgreSQL / SQLite) • API Design (FastAPI / Flask / REST) • CI/CD Discipline (Jenkins, Docker) • Multi-Agent Systems with Schema Contracts • RAG over Regulated Corpora • React / TypeScript Front-End • GenAI Tool Proficiency (Claude / GPT / Ollama) • Compiled-Language Fundamentals (C++) → Go-Ready • F-1 OPT, US-Remote Ready

PROFESSIONAL EXPERIENCE

Progress Solutions Inc. *Jul 2025 - Present*

Jr. AI / ML Engineer Trainee

USA - Healthcare Technology, AI Consulting, SaaS Healthcare

- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility, care navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; agent stability up **~25%**; hallucination reduction **>30%**.
- Built production **RAG systems** for clinical knowledge retrieval and healthcare documentation search; recursive semantic chunking and transformer-based embeddings improved **contextual retrieval precision ~35%** and reduced irrelevant retrieval **>30%**; retrieval-grounded response alignment exceeded **90%**.
- Designed and shipped a **five-agent Healthcare Claims pipeline** (Intake, Validation, Consistency, Duplicate Detection, Risk Scoring) with **schema-validated JSON contracts (pydantic)** between agents to prevent cascading hallucinations - audit-ready reasoning traces for compliance review.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction and care engagement scoring; recall up **15-20%** on high-risk categories with precision held **>90%** via class weighting, stratified sampling, and threshold calibration.
- Packaged ML / LLM inference as **FastAPI / Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; cut post-deployment defects **~30%**.
- Built preprocessing pipelines (Pandas, NumPy) for EHR extracts, appointment histories, and support ticket logs; raised dataset reliability above **98%** and cut downstream model instability **~40%**.
- Maintained HIPAA-conscious data governance: de-identification, data-lineage documentation, evaluation audit trails, and stakeholder system-limitation docs.

Energy Solutions International (Emerson) *Jun 2022 - Apr 2023*

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas, Enterprise ERP, Industrial Automation

- Optimized SQL and **T-SQL stored procedures** powering Synthesis Order-to-Cash; reduced query execution time **~20%** and data retrieval latency **~25%**.
- Built **CI/CD pipelines with Jenkins** for schema updates and stored-procedure deploys; cut deployment errors **>30%** and improved release-cycle time **~35-40%**.
- Designed performance dashboards using SQL Server DMVs and Grafana to track long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence **~25%**.

Suvidha Foundation *Jan 2022 - Mar 2022*

Machine Learning Engineer

Hyderabad, India - Nonprofit Tech, Video Analytics, Applied NLP

- Built a **transformer-based video summarization** system over 5,000+ recorded sessions with hierarchical abstractive summarization and timestamp-anchored clip extraction. Manual review time cut **60-70%**; **~85%** alignment with human-curated highlights.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); cut hallucinations **~30%** and lifted contextual answer accuracy **>85%**.
- Deployed the stack as a Flask-based API with a lightweight web interface for non-technical nonprofit staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Five-stage multi-agent pipeline (Intake, Validation, Consistency, Duplicate, Risk Scoring) with schema-validated JSON contracts between agents preventing cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents. ANN similarity for duplicate detection. Audit-ready explainable risk scoring with reasoning traces.

Python, LangChain, FastAPI, vector retrieval (ANN), pgvector, GPT-4 / Claude, schema contracts (pydantic)

Manga Lens - Multi-Provider AI Vision Chrome Extension (shipped solo, Chrome Web Store)

Real-time AI manga / webtoon translation extension shipped solo to the Chrome Web Store. Multi-provider abstraction across 4 AI vision providers (Claude Sonnet / GPT-4o mini / GPT-4.1 Nano / Ollama) with per-provider payload routing (WebP cloud, JPEG Ollama after CUDA crashes), 7-day translation cache, 29 site selectors via Manifest V3 service workers, narrowed host permissions, privacy policy.

TypeScript, Manifest V3, Service Workers, Claude / GPT-4o mini / GPT-4.1 Nano / Ollama, npm

Dream Decoder - Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app coordinating multimodal APIs (Perplexity Sonar for interpretation, Replicate image models for 16:9 visuals). Intermediate structured-prompt transformation layer lifted contextual alignment ~30% and first-pass image success ~25-30% over naive direct-prompt orchestration.

FastAPI, React + TypeScript + Vite, Tailwind, Perplexity Sonar, Replicate, Docker

Agentic Pixel Character Synthesis & Animation Engine (ongoing)

Phased generative AI system for identity-consistent, pose-controlled pixel character synthesis with sprite-sheet export. Stable Diffusion fine-tuning + LoRA for identity consistency, ControlNet + OpenPose/MediaPipe for pose conditioning, latent-space consistency for animation smoothness, and an LLM-based agent orchestrator that decomposes high-level prompts into generation tasks.

PyTorch, Hugging Face Diffusers, FastAPI, Docker, Stable Diffusion, LoRA, ControlNet

Driver Drowsiness Detection - Real-Time YOLOv8

Replaced a two-stage CNN pipeline with a unified YOLOv8 detect-and-classify model; reduced inference latency ~30%. Sliding-window confidence aggregation cut blink-driven false positives ~25%; adaptive frame skipping and NMS tuning for stable real-time operation.

YOLOv8, OpenCV, PyTorch, LabelImg

EDUCATION

Master of Science, Computer Science - *Kent State University* *Aug 2023 - May 2025*

Focus: Applied Machine Learning, NLP, Database Systems.

Bachelor of Technology, Computer Science - *KL University* *Jun 2019 - May 2023*

Founder / Lead of E-Farming platform (university project, 80-120 active users phase one).

CERTIFICATIONS

Deep Learning Specialization - *DeepLearning.AI (Coursera)* *2024*

SKILLS

Languages: Python, TypeScript, JavaScript, SQL (T-SQL, PostgreSQL, SQLite), C++ (Arduino), PHP

AI / ML: PyTorch, scikit-learn, XGBoost, LangChain, LlamaIndex, Hugging Face Transformers, Diffusers, YOLOv8, OpenCV, ControlNet, Stable Diffusion

LLM / GenAI: RAG, agentic workflows, schema-validated structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding rules, embeddings, vector search (FAISS / pgvector / Pinecone-portable), audit-trace reasoning, multi-agent orchestration, multi-provider AI routing (Claude / GPT-4o / Perplexity / Replicate / Ollama)

Backend: FastAPI, Flask, Node.js (Manifest V3 service workers), REST, Docker, structured logging, load simulation

Frontend: React, Vite, Tailwind, Chrome Extensions (Manifest V3 + content scripts + service workers)

Data: PostgreSQL + pgvector, Pandas, NumPy, ANN similarity, embeddings, SQL data pipelines, T-SQL stored procedures

DevOps: Docker, Jenkins, GitHub Actions, Grafana, AWS/GCP/Azure-portable cloud-ready deployment

Domains: Applied AI, Healthcare AI (HIPAA-conscious), Regulated-Domain LLM Systems, Document Processing + RAG, Multi-Agent Validation, Forward Deploy / Customer-Facing Delivery, Nonprofit ML